ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why one factor is changed.
2	**(d)** greater	12 degrees C versus 7 degrees C.
3	**(b)** 50 degrees C	Subtract zero error of 3.
4	**(c)** A is true but R is false.	A is true; R is false, repetition checks reliability but cannot guarantee truth.
5	**(a)** B and C	Only light differs between B and C.
6	**(d)** plants without fertiliser grown in same conditions	Needed for comparison.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Only testable statements can be supported or rejected.
8	**(b)** water amount is not controlled	Another variable changes.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Equal thickness is a controlled variable; R is the rule of a fair test.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition that explains why A holds.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	T = 34 / 20 = 1.7 s (1); Timing many swings reduces the error of reaction time (1).	1+1
12	Accuracy is closeness to the true value; precision is closeness of repeated readings (1); Readings may be precise but wrong due to zero error (1).	1+1
13	Numbers and units are clear at a glance (1); Comparison and plotting are easier (1).	1+1
14	Sum of five = 62.0 (1); Fifth = 62.0 - 49.4 = 12.6 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Only one soil type was compared (1); The conclusion over-generalises (1); Test several soils with controls (1).	1+1+1
16	Straight line (1); Passes through the origin (1); Ratio stays constant (1).	1+1+1
17	Equal size sheets, same water volume (1); Measure water absorbed (1); Repeat and average (1).	1+1+1
	OR: Outlier: 6.5 s (1); Cause: late stop or a different path (1); Mean without it = 16.4 / 4 = 4.1 s (1).	1+1+1
18	Wanting a certain result can influence readings (1); Use blind or double checks (1); Record all readings honestly (1).	1+1+1
	OR: Experiment changes a variable to see effect (1); Survey collects information without changing anything (1); e.g. fertiliser test versus asking students' favourite fruit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Amount of water in the dough (1). (ii) Time taken to puff (or degree of puffing) (1). (iii) Quantity of flour / thickness of puri / oil temperature (any one) (1). (iv) Hypothesis: puris made with more water puff in less time (or puff better) (1). Repeating trials checks for chance errors and makes the result reliable (1).	1+1+1+2
20	Height increases up to 60 mL (1); Falls at 80 mL (1); Too much water may reduce air in the soil (1); Next test: values between 40 and 80 mL (1); Repeat with several plants (1).	1+1+1+1+1
	OR: Sum = 1500 (1); Mean = 150 cm (1); Range = 158 - 140 = 18 cm (1); Bar graph/histogram suits it (1); Units to be given on axes (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up the **independent** variable (changed by you) with the **dependent** variable (measured).
- Changing more than one factor at a time and still calling the test fair.

- Writing a hypothesis as a question or an opinion instead of a testable statement.
- Taking a single reading and skipping repeats; forgetting to average readings.
- Reading a scale or thermometer from the side (parallax) and omitting units in the answer.
- Drawing a conclusion that the data do not support, or generalising from one trial.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1.6 s	48 / 30 = 1.6 s.
2	**(b)** Does 200 mL of water boil faster than 400 mL on the same burner?	Testable and specific.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Averaging reduces random error.
4	**(d)** 2.1	(2.0+2.2+2.1+2.1)/4 = 2.1.
5	**(d)** check the procedure of that group	Find the cause.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Numbers with units are quantitative.
7	**(d)** temperature and water were not compared	Other factors not controlled.
8	**(c)** it is not meaningful	Unit missing.
9	**(d)** A is false but R is true.	A is false because repeating readings improves reliability; R is true.
10	**(c)** A is true but R is false.	A is true (parallax error); R is false: the eye must be placed directly in line with the mark.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Amount of wax is another variable (1); Use equal amounts of wax (1).	1+1
12	So others can repeat the work (1); Repeating confirms the result (1).	1+1
13	Light promotes growth (1); Only one plant each; repeat with many plants (1).	1+1
14	To compare the effect of the medicine (1); Some improvement may occur without the drug (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Same cloth, same stain, same water (1); Same quantity and time (1); Only the brand is changed (1).	1+1+1
16	A result against expectation is still data (1); It may show the hypothesis is wrong (1); Check the method and use it to improve ideas (1).	1+1+1
17	Small differences look large (1); The scale does not start at zero (1); Always check the axis values (1).	1+1+1
	OR: Independent: size of flame (1); Dependent: time to heat to a fixed temperature (1); Hypothesis: larger flame heats the water faster (1).	1+1+1
18	Others examine the method and data (1); Mistakes and biases are found (1); Reliable findings become accepted knowledge (1).	1+1+1
	OR: Universal understanding (1); Reduce errors in communication (1); Allow comparison of data (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 90 - 78 = 12 degrees C (1). (ii) 69 - 62 = 7 degrees C (1). (iii) A thermometer (1). (iv) No: the fall is 12, 9 and 7 degrees C in successive 2-minute intervals, so cooling slows down as the tea gets closer to room temperature (2).	1+1+1+2
20	Location differs (light, temperature) (1); Plants may differ in type/age (1); Only one plant per condition (1); Improve: same place, many plants, same care (1); Measure growth numerically (1).	1+1+1+1+1
	OR: Hypothesis: testable explanation (1); Law: well-tested description of regularity (1); Opinion: personal view not tested (1); Example for each (1); Only evidence-supported ideas stay in science (1).	1+1+1+1+1

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- Writing a hypothesis as a question or an opinion instead of a testable statement.

- Taking a single reading and skipping repeats; forgetting to average readings.
- Reading a scale or thermometer from the side (parallax) and omitting units in the answer.
- Drawing a conclusion that the data do not support, or generalising from one trial.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** biased and small sample	Weak evidence.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Over-generalising is unsafe.
3	**(d)** measure a pile of many sheets and divide	Reduces error.
4	**(b)** reporting a failed trial	Report everything.
5	**(d)** A is false but R is true.	A is false, R is true.
6	**(b)** boiling at the same temperature for water and salt solution	No difference rejects it.
7	**(d)** unit given in the heading	Standard practice.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Goggles protect the eyes.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but a table's layout does not explain why a graph shows a relationship.
10	**(d)** A is false but R is true.	The deliberately changed quantity is the independent variable, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Independent: length of wire (1); Dependent: resistance (1).	1+1
12	Another experiment may give a different result (1); It can only be supported by evidence (1).	1+1
13	Zero or scale error (1); Correct by +2 degrees C or replace the instrument (1).	1+1
14	A small number of trials (1); Chance can give such runs (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Not yet established by repeated tests (1); Take equal volumes at different temperatures, same freezer (1); Measure times and repeat (1).	1+1+1
16	Counting speed varies (1); A stopwatch gives exact time (1); Allows repeatable measurements (1).	1+1+1
17	Digital thermometers and balances (1); Electronic timers (1); Sensors that log data automatically (1).	1+1+1
	OR: Tilt changes the apparent level (1); Eye in line avoids parallax (1); Reading the bottom of the curve gives true volume (1).	1+1+1
18	Record at fixed intervals (1); Use a table (1); Plot a line graph (1).	1+1+1
	OR: Why does food spoil? led to refrigeration (1); How can we see far? led to telescopes (1); How can we reduce disease? led to vaccines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Length is a controlled variable; changing it would also change the period, so the test would not be fair (1). (ii) 40 / 20 = 2 s (1). (iii) A larger time interval reduces the percentage error due to reaction time (1). (iv) Independent: mass of the bob; dependent: time period (1+1).	1+1+1+2
20	Clear aim (1); Detailed method (1); Data with units (1); Analysis with graphs (1); Honest conclusion with limitations (1).	1+1+1+1+1
	OR: Small sample (1); No control group (1); People may recover on their own (1); Increase the sample and compare with a placebo (1); Independent review of results (1).	1+1+1+1+1

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- Writing a hypothesis as a question or an opinion instead of a testable statement.
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